Structure Checklist for the JML Tutorial

Every topic, named (say) X, should have:

An entry in index.md in this directory (`tutorial`) and an entry in `exercises/exercises.md` (structured the same as `index.md` in this directory), probably of the form:

```
* [X](X)
```

with the first X possibly being an appropriate English phrase for the topic instead of a name without spaces (like the second X).

1. A TUTORIAL SECTION/LESSON

which is a file named `X.md` in this directory (`tutorial`), with the following:

a. starting with a title of the form

```
---
title: JML Tutorial - X
---
```

with the X possibly being a phrase.

b. Some explanatory text (which is the main explanation of X)

c. A second level heading of the form

```
## **[Exercises](https://www.openjml.org/tutorial/exercises/XEx.html)**
```

with a body reading:

```
Follow the link in the above heading to work on the exercises for this lesson.
```

d. (Optionally) a second level heading of the form:

```
## Resources
```

with the body being a list of relevant links, including links to the Java code of any examples used in the text.

2. EXERCISES,

For topic X go in a file named `exercises/XEx.md`

a. starting with a title of the form

```
---
title: JML Tutorial - Exercises - X
---
```

where the X may be an English phrase.

b. A first level heading of the form

```
# X Exercises:
```

c. A second level heading linking back to the tutorial lesson of the form:

```
## [X tutorial](https://www.openjml.org/tutorial/X)
```

d. a numbered list of questions, with each one having:

(i). A second level heading of the form

```
## **Question 1**
```

(Be sure there are no spaces after the first pair of stars (**) or before the second.)

(ii). A boldfaced (**) question, illustrated, if appropriate, by code in a file named (say) `XEx1.java`. The file `XEx1.java` should start with the comment line:

```
// openjml --esc XEx1.java
```

(iii). If the question is about an entire class or interface, use the following to include the source file.

```
```Java
{% include_relative XEx1.java %}
```
```

(iv). If the file `XEx1.java` has any verification errors, then add a file named (say) `XEx1.out`, with any verification errors indicated by a comment at the end of the line of the form: `// ERROR: ...`

(with a reason that doesn't make the question trivial.)

e. a paragraph (or list) starting with

```
**Learning Objectives:**
```

f. A second-level heading of the form

```
## **[Answer Key](XExKey)**
```

g. A second-level heading of the form

```
## Resources
```

followed by a list of relevant resources, including links to any Java code of the form

```
+ [XEx1 Java code](XEx1.java)
```

3. An ANSWER KEY,

This should be in a file named `exercises/XExKey.md` with:

a. a title of the form

```
---
title: JML Tutorial - Exercises – X
---
```

where X may be a phrase

b. a first level heading of the form:

```
# X Exercises Key:
```

c. a second level heading linking to the tutorial section of the form:

```
## [X tutorial](https://www.openjml.org/tutorial/X)
```

d. a numbered list of answers to the questions, with each having:

(i). An introductory sentence or paragraph,

(ii). A Java file (with JML) that solves the exercise, named (say) `Xex1Ans.java`. Paste the contents of that file into `XExKey.md` between the following lines

```
```Java
```
```

but with the first line (the comment to run OpenJML) removed and with all occurrences of `XEx1Ans` replaced everywhere by `XEx1`

(iii). The `XEx1Ans.java` file should verify, but if it does not, there should be an `XEx1Ans.out` file created by executing `make XEx1Ans.out`

(iv). Any additional explanation or thoughts on the question or solution, possibly showing alternatives

e. a second level heading of the form:

```
## Resources
```

followed by a list including any relevant resources, including at least the following:

```
+ [X Exercises](XEx)
+ [All exercises](exercises)
```